# §1 绪论

## §1.4 不确定关系

$$\Delta x = \sqrt{\langle (\hat{x} - \langle \hat{x} \rangle)^2 \rangle} = \sqrt{\langle \hat{x}^2 \rangle - \langle x \rangle^2}$$

$$\Delta p = \sqrt{\langle \hat{p}^2 \rangle - \langle p \rangle^2}$$

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

例：设归一 Gaussian 波包 $\psi(x) = A e^{ikx} e^{-\frac{x^2}{2\sigma^2}}$ 描述，计算 $\Delta x, \Delta p$

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi^*(x)\, x\, \psi(x)\, dx$$

$$= A^2 \int_{-\infty}^{\infty} x e^{-\frac{x^2}{\sigma^2}} dx = \frac{1}{2} A^2 \sigma^2 e^{-\frac{x^2}{\sigma^2}} \Big|_{-\infty}^{\infty} = 0$$

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} \psi^*(x)\, x^2\, \psi(x)\, dx$$

$$= A^2 \int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{\sigma^2}} dx$$

$$= \frac{1}{2} A^2 (-\sigma^2) \int_{-\infty}^{\infty} x\, d\left(e^{-\frac{x^2}{\sigma^2}}\right) = -\frac{1}{2} A^2 \sigma^2 \left( x e^{-\frac{x^2}{\sigma^2}} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} e^{-\frac{x^2}{\sigma^2}} dx \right)$$

$$= \frac{1}{2} A^2 \sigma^2 \cdot \sqrt{\pi \sigma^2} = \frac{1}{2} \sqrt{\pi}\, A^2 \sigma^3$$

$$I = \int_{-\infty}^{+\infty} e^{-ax^2} dx$$

$$I^2 = \int_{-\infty}^{+\infty} e^{-ax^2} dx \int_{-\infty}^{+\infty} e^{-ay^2} dy$$

$$= \iint e^{-ar^2} r\, dr\, d\theta$$

$$= \frac{1}{2a} \cdot 1 \cdot 2\pi = \frac{\pi}{a} \quad \Rightarrow \quad I = \sqrt{\frac{\pi}{a}}$$

$$\Rightarrow \Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \frac{\pi^{1/4}}{\sqrt{2}} \sqrt{\sigma}\, A^{3/2}$$

$$\langle \hat{p} \rangle = \int_{-\infty}^{+\infty} \psi^*(x) (-i\hbar \nabla) \psi(x)\, dx$$

$$= (-i\hbar) A^2 \int_{-\infty}^{+\infty} e^{-ikx} e^{-\frac{x^2}{2\sigma^2}} \nabla \left( e^{ikx} e^{-\frac{x^2}{2\sigma^2}} \right) dx$$

$$= (-i\hbar) A^2 \int_{-\infty}^{+\infty} e^{-ikx} e^{-\frac{x^2}{2\sigma^2}} e^{ikx} \left( ik e^{-\frac{x^2}{2\sigma^2}} - \frac{2x}{2\sigma^2} e^{-\frac{x^2}{2\sigma^2}} \right) dx$$

$$= (-i\hbar) A^2 \int_{-\infty}^{\infty} \left( ik - \frac{x}{\sigma^2} \right) e^{-\frac{x^2}{\sigma^2}} dx$$

$$= (-i\hbar) A^2 \cdot ik \cdot \sqrt{\pi}\, \sigma = \sqrt{\pi}\, k \hbar A^2 \sigma$$

$$\langle \vec{p}^2 \rangle = \int_{-\infty}^{\infty} \psi^*(x)\, \vec{p}^2\, \psi(x)\, dx$$

$$= -\hbar^2 \alpha^2 \int_{-\infty}^{\infty} e^{-ik_0 x} e^{-\frac{\alpha^2 x^2}{2}} \frac{d^2}{dx^2}\left(e^{ik_0 x} e^{-\frac{\alpha^2 x^2}{2}}\right) dx$$

$$= -\hbar^2 \alpha^2 \int_{-\infty}^{\infty} e^{-ik_0 x} e^{-\frac{\alpha^2 x^2}{2}} \frac{d}{dx}\left[ e^{ik_0 x} e^{-\frac{\alpha^2 x^2}{2}} \left(ik_0 - \frac{\alpha^2 x}{2}\right)\right] dx$$

$$= -\hbar^2 \alpha^2 \int_{-\infty}^{\infty} e^{-ik_0 x} e^{-\frac{\alpha^2 x^2}{2}} e^{ik_0 x} e^{-\frac{\alpha^2 x^2}{2}} \left(ik_0 - \frac{\alpha^2 x}{2}\right)^2 dx$$

$$= -\hbar^2 \alpha^2 \int_{-\infty}^{\infty} e^{-\alpha^2 x^2} \left(ik_0 - \frac{\alpha^2 x}{2}\right)^2 dx$$

$$= -\hbar^2 \alpha^2 \int_{-\infty}^{+\infty} \left(-k_0^2 - \frac{\alpha^4}{4} x^2 + \frac{ik_0 \alpha^2}{2} x + \frac{\alpha^2}{2} x\right) e^{-\frac{\alpha^2}{2} x^2} dx$$

$$\int_{-\infty}^{\infty} x^2 e^{-\frac{\alpha^2}{2} x^2} dx = 2 \int_0^{\infty} x^2 e^{-\frac{\alpha^2}{2} x^2} dx = \langle (\Delta x)^2 \rangle$$

## §1.3 Schrodinger 方程

经典力学：$E = \frac{\vec{p}^2}{2m} + V$，$E \to i\hbar\frac{\partial}{\partial t}$，$\vec{p} \to -i\hbar\nabla$

$$i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r},t)\right]\psi(\vec{r},t)$$

1. 几率流、几率守恒 连续性方程

$$\frac{\partial}{\partial t}\rho(\vec{r},t) + \nabla\cdot\vec{J}(\vec{r},t) = 0 \quad \text{其中 } \rho(\vec{r},t) := \psi^*(\vec{r},t)\psi(\vec{r},t)$$

$$\begin{cases} i\hbar\frac{\partial}{\partial t}\psi(\vec{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r},t)\right]\psi(\vec{r},t) \\ -i\hbar\frac{\partial}{\partial t}\psi^*(\vec{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r},t)\right]\psi^*(\vec{r},t) \end{cases} \Leftarrow \text{复共轭}$$

$\cdot\psi^*$

$\cdot\psi$

$$\text{则 } i\hbar\frac{\partial}{\partial t}\left[\psi^*(\vec{r},t)\psi(\vec{r},t)\right] = \left(-\frac{\hbar^2}{2m}\right)\left[\psi^*(\vec{r},t)\nabla^2\psi(\vec{r},t) - \psi(\vec{r},t)\nabla^2\psi^*(\vec{r},t)\right]$$

$$= -\frac{\hbar^2}{2m}\nabla\cdot(\psi^*\nabla\psi - \psi\nabla\psi^*)$$

$$\text{令 } \vec{J} = \frac{\hbar}{2mi}(\psi^*\nabla\psi - \psi\nabla\psi^*)$$

$$= \frac{\hbar}{m}\operatorname{Im}(\psi^*\nabla\psi)$$

$$\text{引入相位 } \psi(\vec{r},t) = \sqrt{\rho(\vec{r},t)}\,e^{\frac{i\theta(\vec{r},t)}{\hbar}}$$

$$\text{则 } \vec{J} = \frac{\hbar}{m}\operatorname{Im}\left[\sqrt{\rho}\,e^{\frac{i\theta}{\hbar}}\nabla\left(\sqrt{\rho}\,e^{-\frac{i\theta}{\hbar}}\right)\right]$$

$$= \frac{\hbar}{m}\operatorname{Im}\left[\sqrt{\rho}\,e^{\frac{i\theta}{\hbar}}\left(\frac{\nabla\rho}{2\sqrt{\rho}} + \sqrt{\rho}\,\frac{i}{\hbar}\nabla\theta\right)e^{-\frac{i\theta}{\hbar}}\right]$$

$$= \frac{\hbar}{m}\operatorname{Im}\left(\rho\,\frac{i}{\hbar}\nabla\theta\right)$$

$$= \frac{\rho}{m}\nabla\theta$$

2. 定态 Schrodinger 方程

分离变量：$\psi(\vec{r},t) = \psi(\vec{r})f(t)$

$$\text{则 } E_n = \frac{i\hbar\frac{\partial f}{\partial t}}{f(t)} = \frac{1}{\psi(\vec{r})}\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r})\right]$$

解得 $f(t) = e^{-\frac{iE_nt}{\hbar}}$

能量本征态 $\{\psi_n\}$

$$\phi(\vec{r},t) = \sum_n \phi_n\psi_n(\vec{r})e^{-\frac{iE_nt}{\hbar}}$$

$$\text{其中 } \phi_n = \int d^3r\,\psi_n^*(\vec{r})\phi(\vec{r},0)$$

# §1.6 一维定态问题

## 1. 无限深势井

$$V(x)=\begin{cases}0 & -\frac{a}{2}<x<\frac{a}{2}\\ \infty & x<-\frac{a}{2}\text{ 或 }x>\frac{a}{2}\end{cases}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}=E\psi \Rightarrow \psi=Ae^{ikx}+Be^{-ikx},\quad k=\frac{\sqrt{2mE}}{\hbar}$$

$$\psi\left(\frac{a}{2}\right)=\psi\left(-\frac{a}{2}\right)=0 \Rightarrow \begin{cases}Ae^{\frac{ika}{2}}+Be^{-\frac{ika}{2}}=0\\ Ae^{-\frac{ika}{2}}+Be^{\frac{ika}{2}}=0\end{cases}$$

非平凡解要求：$\begin{vmatrix} e^{\frac{ika}{2}} & e^{-\frac{ika}{2}} \\ e^{-\frac{ika}{2}} & e^{\frac{ika}{2}}\end{vmatrix}=0 \Rightarrow \sin ka=0,\ ka=n\pi$

此时 $Ae^{\frac{in\pi}{2}}+Be^{-\frac{in\pi}{2}}=e^{\frac{in\pi}{2}}\left(A+Be^{-in\pi}\right)=e^{i\frac{n\pi}{2}}(A+B)=0 \rightarrow A=-B$

$$\psi(x)=A\left(e^{ikx}-e^{-ikx}\right)=2iA\sin\left(\frac{n\pi}{a}x\right)$$

$$=2iAe^{-i\frac{n\pi}{2}}\sin\frac{n\pi}{a}\left(x+\frac{a}{2}\right),\quad n=1,2,\cdots$$

归一化条例：$\int_{-\infty}^{\infty}|\psi(x)|^2dx$

$$=\int_{-\frac{a}{2}}^{\frac{a}{2}}\psi A^2\sin^2\frac{n\pi}{a}\left(x+\frac{a}{2}\right)dx$$

$$=\psi A^2\cdot\frac{a}{2}=2A^2a=1 \rightarrow A=\sqrt{\frac{1}{2a}}$$

$$\psi_n(x)=\sqrt{\frac{2}{a}}\sin\frac{n\pi}{a}\left(x+\frac{a}{2}\right)$$ 注意：势阱宽度为 $a$，以 $x=0$ 为中心。

$$E_n(x)=\frac{\hbar^2k^2}{2m}=\frac{n^2\pi^2\hbar^2}{2ma^2}$$

正交性：$\int_{-\frac{a}{2}}^{\frac{a}{2}}\psi_m(x)\psi_n(x)dx=\delta_{mn}$

* 四界条例：

① 一般 $\psi(x)$，$\frac{d\psi}{dx}$ 连续

② $V(x)$ 在四界发散，$\frac{d\psi}{dx}$ 有不连续

例：$\delta$ 函数势的边界条件

$V(x) = \alpha\delta(x)$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \alpha\delta(x)\psi(x) = E\psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} = \frac{2m}{\hbar^2}\left[\alpha\delta(x) - E\right]\psi(x)$$

$$\frac{d\psi(0^+)}{dx} - \frac{d\psi(0^-)}{dx} = \frac{2m}{\hbar^2}\lim_{\eta\to 0}\int_{-\eta}^{\eta}\left[\alpha\delta(x) - E\right]\psi(x)\,dx \qquad \int_{-\infty}^{\infty}\delta(x)\,dx = 1$$

$$= \frac{2m\alpha}{\hbar^2}\psi(0)$$

$$\psi(0^-) = \psi(0^+)$$

③ $V(x)$ 为边界不发散，但有跃变，$\frac{d\psi}{dx}$ 连续.

$$\frac{d^2\psi(x)}{dx^2} = \frac{2m}{\hbar^2}(V - E)\psi(x)$$

$$\rightarrow \frac{d\psi(x_0+0^+)}{dx} - \frac{d\psi(x_0-0^-)}{dx} = \frac{2m}{\hbar^2}\lim_{\eta\to 0}\int_{x_0-\eta}^{x_0+\eta}(V-E)\psi\,dx = 0$$

有限.

2. 自由粒子 $V(x) = 0$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \rightarrow \psi(x) = Ae^{ikx}, \quad k = \sqrt{\frac{2mE}{\hbar^2}} = \frac{p}{\hbar}$$

(1) $\delta$ 函数的一些性质：$\delta(x) = \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases}$

$$\int_{-\infty}^{\infty}\psi_k^*(x)\psi_{k'}(x)\,dx = A^2\int_{-\infty}^{\infty}e^{-i(k-k')x}\,dx = 2\pi A^2\delta(k-k')$$

取 $A = \frac{1}{\sqrt{2\pi}}$，$\psi_k(x) = \frac{1}{\sqrt{2\pi}}e^{ikx}$.

$$\int_{-\infty}^{\infty}\psi_k^*(x)\psi_{k'}(x)\,dx = \delta(k-k')$$

类似 $C_i a_j = \delta_{ij}$

$\delta[\alpha(x-x_0)] = \frac{1}{|\alpha|}\delta(x-x_0)$

$f(x)\delta(x-x_0) = f(x_0)\delta(x-x_0)$

$\int_{-\infty}^{\infty}dx\,\delta(x-x_0)\delta(x-x_1) = \delta(x_1-x_0)$

$\int_{-\infty}^{\infty}dx\left[\frac{x^n}{n!}\delta(x-x_0)\right]f(x) = f^{(n)}(x_0)$

$\delta(x-x_0)\frac{x^n}{n!} = \left[\delta(x-x_0)\right]\frac{x^n}{n!} = \frac{1}{n!}\delta(x-x_0)x^n$

$\delta(x-x_0)\frac{x^n}{n!} - \delta(x-x_0)\frac{x_0^n}{n!} = \delta(x-x_0)(x-x_0)$

(2) 箱归一化：$[-\frac{L}{2}, \frac{L}{2}]$

周期性边界条件：$\psi(-\frac{L}{2}) = \psi(\frac{L}{2})$

$\Rightarrow e^{-\frac{ikL}{2}} = e^{\frac{ikL}{2}}$，$e^{ikL} = 1$，$\frac{kL}{2\pi} = n$

$p_n = \frac{2\pi\hbar}{L}n$，$\psi_n(x) = Ae^{\frac{ip_nx}{\hbar}} = Ae^{\frac{2\pi i}{L}nx}$

正交：$\int_{-\frac{L}{2}}^{\frac{L}{2}}\psi_m^*(x)\psi_n(x)\,dx = \int_{-\frac{L}{2}}^{\frac{L}{2}}A^2e^{2\pi i\frac{(n-m)x}{L}}\,dx$

$= A^2 \delta_{mn}$

$L \to \infty$ 时，动量取值为连续值。

3. 有限深方势井

$$V(x) = \begin{cases} -V_0 & -\frac{a}{2} < x < \frac{a}{2} \\ 0 & x < -\frac{a}{2},\ x > \frac{a}{2} \end{cases}$$

(1) 束缚态，$E < V(+\infty) = 0$ 且 $\psi(\pm\infty) \to 0$

① $x < -\frac{a}{2}$：

$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \to \psi(x) = Ae^{-\alpha x} + Be^{\alpha x}$，其中 $\alpha = \sqrt{\frac{-2mE}{\hbar}}$

$\psi(x) = Be^{\alpha x}$ $\quad \psi(-\infty) \to 0$，$A = 0$

② $-\frac{a}{2} < x < \frac{a}{2}$：

$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = (E - V_0)\psi \to \psi(x) = C(e^{ikx} \pm e^{-ikx})$，其中 $k = \sqrt{\frac{2m(E-V_0)}{\hbar}}$

束缚态为确定的宇称（对称的）

③ $x > \frac{a}{2}$：

$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \to \psi(x) = De^{-\alpha x} + Fe^{\alpha x}$ $\quad \psi(+\infty) \to 0$，$F = 0$

$\psi(x) = De^{-\alpha x}$

偶宇称：$B = D$，+

解为 $\begin{cases} Be^{\alpha x} & x < -\frac{a}{2} \\ C(e^{ikx} + e^{-ikx}) = 2C\cos kx & \\ Be^{-\alpha x} & x > \frac{a}{2} \end{cases}$

边界 $\begin{cases} Be^{-\alpha\frac{a}{2}} = 2C\cos(k\frac{a}{2}) \\ \alpha Be^{-\alpha\frac{a}{2}} = 2Ck\sin(k\frac{a}{2}) \end{cases}$

B、C 的非平凡解要求：

$$\begin{vmatrix} e^{-\alpha\frac{a}{2}} & 2\cos(\frac{ka}{2}) \\ \alpha e^{-\alpha\frac{a}{2}} & 2k\sin(\frac{ka}{2}) \end{vmatrix} = e^{-\alpha\frac{a}{2}} 2k\sin\frac{ka}{2} - 2\cos\frac{ka}{2}\alpha e^{-\alpha\frac{a}{2}}$$

$$= 2e^{-\alpha\frac{a}{2}}\cos\frac{ka}{2}\left[k\tan(\frac{ka}{2}) - \alpha\right] = 0$$

解得 $\alpha = k\tan(\frac{1}{2}ka)$ （偶宇称）

奇宇称：$B=-D$，

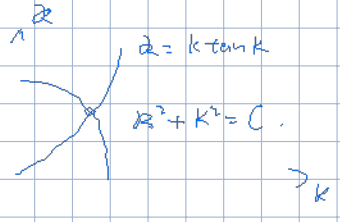
$$\text{解}\begin{cases} Be^{\alpha x} & x<-\frac{a}{2} \\ C(e^{ikx}-e^{-ikx}) = 2Ci\sin kx & \\ -Be^{-\alpha x} & x>\frac{a}{2} \end{cases}$$

$$\text{边界}\begin{cases} Be^{-k\frac{a}{2}} = 2Ci\sin(-\frac{1}{2}ka) \\ \alpha Be^{-k\frac{a}{2}} = 2kCi\cos(-\frac{1}{2}ka) \end{cases}$$

$$\begin{vmatrix} e^{-k\frac{a}{2}} & 2\sin(-\frac{1}{2}ka) \\ \alpha e^{-k\frac{a}{2}} & 2k\cos(-\frac{1}{2}ka) \end{vmatrix} = e^{-k\frac{a}{2}}\left[\cos(\tfrac{1}{2}ka)\left|k+\alpha\tan(\tfrac{1}{2}ka)\right|\right]=0$$

解得 $\alpha=-k\cot(\frac{1}{2}ka)$ （奇宇称）

$\alpha=\sqrt{\frac{-2mE}{\hbar}}$，$k=\sqrt{\frac{2m(E-V_0)}{\hbar}}$　　图解法：




↳ $k^2+\alpha^2=-\frac{2mV_0}{\hbar}$

一维束缚态：

(1) 束缚能量不简并　假设存在能量相同的态 $\psi_1$、$\psi_2$

设：$E_1=E_2=E$，$\frac{1}{\psi_1}\frac{d^2\psi_1}{dx^2}=\frac{2m}{\hbar}[V(x)-E]=\frac{1}{\psi_2}\frac{d^2\psi_2}{dx^2}$

↳ $\frac{1}{\psi_1}\frac{d\psi_1}{dx}-\frac{1}{\psi_2}\frac{d\psi_2}{dx}=C$

取 $x\to\infty$，$\psi_1(\infty)=\psi_2(\infty)=0$，$C=0$

⇒ $\frac{1}{\psi_1}\frac{d\psi_1}{dx}=\frac{1}{\psi_2}\frac{d\psi_2}{dx}$，解得 $\ln\psi_1=\ln\psi_2+C$，$\psi_1=\lambda\psi_2$

由归一化条件可确定 $\lambda=1$，$\psi_1$、$\psi_2$ 为同一个态

(2) 第 $n$ 个束缚态有 $n-1$ 个节点

引理：束缚态的 $\psi(x)$ 为实函数　非一也可以吧，只要满足定态一方程

$$\frac{1}{\psi_1}\left(\frac{\hbar^2}{2m}\frac{d^2\psi_1}{dx^2}\right)+E_1=V(x)=\frac{1}{\psi_2}\frac{d^2\psi_2}{dx^2}+E_2 \quad [E_1<E_2]$$

↳ $\frac{\hbar^2}{2m}\frac{d}{dx}\left(\psi_1\frac{d\psi_2}{dx}-\psi_2\frac{d\psi_1}{dx}\right)=(E_2-E_1)\psi_1\psi_2$

(a) 命题：每两个连续 $\psi_n(x)$ 节点之间必有一个 $\psi_{n+1}(x)$ 的节点。

假设 $x=a$、$x=b$ 是 $\psi_1(x)$ 的两个连续节点.

$$\int_a^b \text{LHS} = \frac{\hbar^2}{2m}\left[\psi_1(a)\frac{d\psi_2}{dx}(a) - \psi_2(b)\frac{d\psi_1}{dx}(b)\right] = (E_2 - E_1)\int_a^b \psi_1(x)\psi_2(x)\,dx$$

(2) 散射态, $E > U(+\infty) = 0$

① $x < -\frac{a}{2}$, $\psi(x) = Ae^{ikx} + Be^{-ikx}$, $k = \frac{\sqrt{2mE}}{\hbar}$

② $-\frac{a}{2} < x < \frac{a}{2}$, $\psi(x) = Ce^{ikx} + De^{-ikx}$, $k = \frac{\sqrt{2m(E+U_0)}}{\hbar}$

③ $x > \frac{a}{2}$, $\psi(x) = Fe^{ikx}$

由界
$$\begin{cases} Ae^{-ik\frac{a}{2}} + Be^{ik\frac{a}{2}} = Ce^{-ik\frac{a}{2}} + De^{ik\frac{a}{2}} \\ ik\left(Ae^{-ik\frac{a}{2}} - Be^{ik\frac{a}{2}}\right) = ik\left(Ce^{-ik\frac{a}{2}} - De^{ik\frac{a}{2}}\right) \\ Ce^{ik\frac{a}{2}} + De^{-ik\frac{a}{2}} = Fe^{ik\frac{a}{2}} \\ ik\left(Ce^{-ik\frac{a}{2}} - De^{ik\frac{a}{2}}\right) = ik\,Fe^{ik\frac{a}{2}} \end{cases}$$

解得
$$B = i\frac{\sin ka}{2hk}(k^2 - h^2)F$$
$$F = \frac{e^{-ika}A}{\cos(ka) - i\frac{\sin ka}{2hk}(h^2+k^2)}$$

透射系数 $T := \left|\frac{J_F}{J_A}\right|$ $\psi_F = Fe^{ikx}$, $J_F = \frac{F^2}{m}e^{ikx}\frac{\partial}{\partial x}(-i\hbar kx)$

$\psi_A = Ae^{ikx}$, $J_A = \frac{A^2}{m}e^{ikx}\frac{\partial}{\partial x}(-i\hbar kx)$

$$T = \left|\frac{F}{A}\right|^2 = \frac{1}{\cos^2(ka) + \sin^2(ka)\frac{(h^2+k^2)^2}{4k^2h^2}} = \frac{1}{1 + \sin^2(ka)\frac{(k^2-h^2)^2}{4k^2h^2}}$$

$$= \frac{4k^2h^2}{4k^2h^2 + (k^2-h^2)^2\sin^2(ka)} = \frac{\frac{4}{\hbar^2}(2m)^2E(E+V_0)}{\frac{4}{\hbar^2}(2m)^2E(E+V_0) + \left(\frac{2m}{\hbar^2}V_0\right)^2\sin^2(ka)}$$

$$= \frac{4E(E+V_0)}{4E(E+V_0) + V_0^2\sin^2\left(\frac{\sqrt{2m(E+V_0)}}{\hbar}a\right)}$$

反射系数 $R := \left|\frac{J_R}{J_A}\right|$ $\psi_R = Be^{-ikx}$, $J_R = \frac{B^2}{m}e^{-ikx}\frac{\partial}{\partial x}(i\hbar kx)$

$\psi_A = Ae^{ikx}$, $J_A = \frac{A^2}{m}e^{ikx}\frac{\partial}{\partial x}(-i\hbar kx)$

$$= \left|\frac{B}{A}\right|^2 = \left|\frac{F}{A}\right|^2\cdot\left|\frac{B}{F}\right|^2 = T\cdot\left|\frac{\sin ka}{2hk}(k^2-h^2)\right|^2 = \frac{\sin^2(ka)(k^2-h^2)^2}{4k^2h^2 + (k^2-h^2)^2\sin^2(ka)}$$

$$= \frac{V_0^2\sin^2\left(\frac{\sqrt{2m(E+V_0)}}{\hbar}a\right)}{4E(E+V_0) + V_0^2\sin^2\left(\frac{\sqrt{2m(E+V_0)}}{\hbar}a\right)}$$

$T + R = 1$

$$4E(E+V_0)+V_0^2\sin^2\left(\frac{2a\sqrt{2m(E+V_0)}}{\hbar}\right)$$

共振透射：$\frac{a}{\hbar}\sqrt{2m(E+V_0)}=n\pi$

此时 $R=0$

4. 方势井 $V(x)=\begin{cases}V_0 & -\frac{a}{2}\le x\le\frac{a}{2}\\ 0 & \text{其它}\end{cases}$

$$\psi(x)=\begin{cases}Ae^{ikx}+Be^{-ikx} & x<-\frac{a}{2}\\ Ce^{\alpha x}+De^{-\alpha x} & -\frac{a}{2}<x<\frac{a}{2}\\ Fe^{ikx} & x>\frac{a}{2}\end{cases}\qquad k=\frac{\sqrt{2mE}}{\hbar},\quad \alpha=\frac{\sqrt{2m(V_0-E)}}{\hbar}$$

边界
$$\begin{cases}Ae^{-ik\frac{a}{2}}+Be^{ik\frac{a}{2}}=Ce^{-\alpha\frac{a}{2}}+De^{\alpha\frac{a}{2}}\\ ik\left(Ae^{-ik\frac{a}{2}}-Be^{ik\frac{a}{2}}\right)=\alpha\left(Ce^{-\alpha\frac{a}{2}}-De^{\alpha\frac{a}{2}}\right)\\ Fe^{ik\frac{a}{2}}=Ce^{-\alpha\frac{a}{2}}+De^{\alpha\frac{a}{2}}\\ ikFe^{ik\frac{a}{2}}=\alpha\left(Ce^{\alpha\frac{a}{2}}-De^{-\alpha\frac{a}{2}}\right)\end{cases}$$

解得

反射系数 $R=\dfrac{(k^2+\alpha^2)\sinh^2\alpha a}{(k^2+\alpha^2)\sinh^2\alpha a+4k^2\alpha^2}$

透射系数 $T=\dfrac{4k^2\alpha^2}{(k^2+\alpha^2)\sinh^2\alpha a+4k^2\alpha^2}$ 即使 $V_0>0$，穿透概率也不为0

3. δ函数势 $V(x) = \gamma\delta(x)$

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) + \gamma\delta(x)\psi(x) = E\psi(x)$$

$x=0$ 处，一阶导数发散，一阶导数不连续

$$\lim_{\eta\to 0}\int_{-\eta}^{+\eta} \quad \left(\lim_{\eta\to 0}\int_{-\eta}^{+\eta}\delta(x)=1\right)$$

$$\Rightarrow \frac{d}{dx}\psi(0^+) - \frac{d}{dx}\psi(0^-) = \frac{2m\gamma}{\hbar^2}\psi(0)$$

(1) 束缚态 $E<0$

$x\neq 0$, $\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0$, $k = \frac{\sqrt{2mE}}{\hbar}$

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} \\ Ce^{ikx} \end{cases}$$ 粒子从左边入射.

边界条件: $\begin{cases} A+B=C \\ ikC - ik(A-B) = \frac{2m\gamma}{\hbar^2}C \end{cases}$ 解得 $\begin{cases} B = \frac{m\gamma}{i\hbar^2k - m\gamma}A \\ C = \frac{i\hbar^2k}{i\hbar^2k-m\gamma}A \end{cases}$

$$T = \left|\frac{J_t}{J_i}\right| = \left|\frac{C}{A}\right|^2 = \frac{(\hbar^2k)^2}{(\hbar^2k)^2+(m\gamma)^2} = \frac{2mE\hbar^2}{2mE\hbar^2+m^2\gamma^2} = \frac{2\hbar^2E}{2\hbar^2E+m\gamma^2}$$

$$T = \left|\frac{J_r}{J_i}\right| = \left|\frac{B}{A}\right|^2 = \frac{(m\gamma)^2}{(\hbar^2k)^2+(m\gamma)^2} = \frac{m^2\gamma^2}{2mE\hbar^2+m^2\gamma^2} = \frac{m\gamma^2}{2\hbar^2E+m\gamma^2}$$

(2) 束缚态 $E<0$

$x\neq 0$: $\frac{d^2\psi}{dx^2} - \alpha^2\psi = 0$, $\alpha = \frac{\sqrt{-2mE}}{\hbar}$

通解: $\psi(x) = Ae^{\alpha x} + Be^{-\alpha x}$

解为 $\psi(x) = \begin{cases} Ae^{\alpha x} & x<0 \\ Be^{-\alpha x} & x>0 \end{cases}$ 束缚态 $\psi(x)\to 0$

边界条件: $\begin{cases} A=B \\ -\alpha A - \alpha B = -\frac{2m\gamma}{\hbar^2}A \end{cases}$

解得 $\alpha = \frac{m\gamma}{\hbar^2}$, $E = -\frac{m\gamma^2}{2\hbar^2}$

$$= \frac{2\pi^2 mc^2}{2\pi^2 V^2} \int_0^{\infty} x^2 \left[ \sin^2\left(\frac{x}{2}\right) \sin^2\left(\frac{x}{2}\right) \right] dx$$

$$= \frac{2\pi^2 mc^2}{2\pi^2 V^2}$$

$$\langle x \rangle = \frac{V}{2\pi^2} \int_0^{\infty} x^2 \left[ \frac{1}{2} \sin\left(\frac{x}{2}\right) e^{\frac{ix}{\lambda_1 - \lambda_2}} + \frac{1}{2} \sin\left(\frac{x}{2}\right) e^{-\frac{ix}{\lambda_1 - \lambda_2}} \right] \times \left[ \frac{1}{2} \sin\left(\frac{x}{2}\right) e^{\frac{ix}{\lambda_1 - \lambda_2}} + \frac{1}{2} \sin\left(\frac{x}{2}\right) e^{-\frac{ix}{\lambda_1 - \lambda_2}} \right] dx \quad (2)$$

$$= \frac{V}{2\pi^2} \int_0^{\infty} x^2 \left[ \frac{1}{4} \sin^2\left(\frac{x}{2}\right) + \frac{1}{4} \sin^2\left(\frac{x}{2}\right) + \frac{1}{2} \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) \sin\left(\frac{x}{\lambda_2 - \lambda_1}\right) \right] dx$$

$$= \cdots =$$

3. (1)

$$\begin{cases} -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi = E\psi \\ \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V_0\right)\psi = E\psi \end{cases}$$

$$\begin{cases} Ae^{ikx} + Be^{-ikx}, & 0 \le x < a \quad k = \frac{\sqrt{2mE}}{\hbar} \\ Ce^{-Kx}, & a \le x, \quad K = \frac{\sqrt{2m(V_0 - E)}}{\hbar} \end{cases}$$ 束缚态，$V_0 > E$

边界：

$$\begin{cases} A + B = 0 \\ Ae^{ika} + Be^{-ika} = Ce^{-Ka} \\ k\left(Ae^{ika} - Be^{-ika}\right) = -KCe^{-Ka} \end{cases}$$

(2)

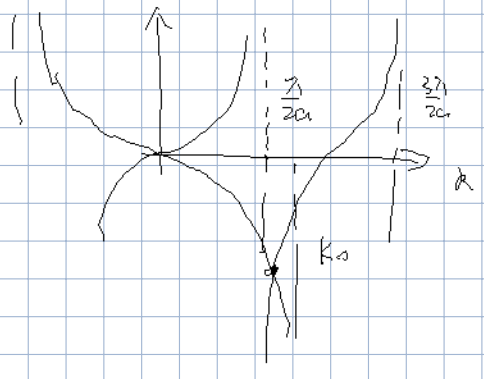
$$\begin{cases} 2iA\sin ka - Ce^{-Ka} = 0 \\ 2ikA\cos ka + kCe^{-Ka} = 0 \end{cases}$$

非零解：

$$\begin{vmatrix} 2i\sin ka & -e^{-Ka} \\ 2ik\cos ka & -ke^{-Ka} \end{vmatrix} = 2e^{ika}\left(K\sin ka + k\cos ka\right) = 0$$

$$\Rightarrow \tan(ka) = -\frac{k}{K}$$

$$\tan\left(\frac{\sqrt{2m(V_0 - E)}}{\hbar}a\right) = -\sqrt{\frac{V_0 - E}{E}}$$




(3) 上述方程仅有一解，

$$k^2 + K^2 = \frac{2mV_0}{\hbar} = k_0^2$$

$$\tan(ka) = -\frac{k}{\sqrt{k_0^2 - k^2}}$$

$$k_0 \in \left(\frac{\pi}{2a}, \frac{3\pi}{2a}\right)$$

$$V_0 \in \left[\frac{\hbar^2\pi^2}{8a^2m}, \frac{9\hbar^2\pi^2}{8a^2m}\right)$$

6. (1) $\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} \\ Ce^{ikx} \end{cases}$ $k = \frac{\sqrt{2mE}}{\hbar}$

$k_2 = \frac{\sqrt{2m(E-V_0)}}{\hbar}$

$\begin{cases} A + B = C \\ k(A - B) = k_2 C \end{cases} \rightarrow$ $B = \frac{k - k_2}{k + k_2} A$

$C = \frac{2k}{k + k_2} A$

$R = \left|\frac{B}{A}\right|^2$, $T = \left|\frac{C}{A}\right|^2 \cdot \frac{k_2}{k}$

(2) $\begin{cases} A + B = C \\ \left.\frac{d\psi}{dx}\right|_{x=0^+} - \left.\frac{d\psi}{dx}\right|_{x=0^-} = \frac{2m}{\hbar^2}\left[-i\gamma(\psi(0))\right] \end{cases}$

$B = \frac{k - k_2 - \beta}{k + k_2 + \beta} A$, $\beta = \frac{2\gamma}{\hbar^2}$

$C = \frac{2k}{k + k_2 + \beta} A$

$k_2(C - k(A - B)) = \frac{2m}{\hbar^2}(-i\gamma C)$

$R = \left|\frac{B}{A}\right|^2 = \frac{(k - k_2 - \beta)^2}{(k + k_2 + \beta)^2}$

$J = \frac{\hbar}{m} Im\{\psi^* \partial\psi\} = \frac{\hbar}{m}|\psi|^2 ik$

$T = \left|\frac{C}{A}\right|^2 \cdot \frac{k_2}{k} = \frac{4k^2}{(k + k_2 + \beta)^2} \cdot \frac{k_2}{k}$

$A = 1 - R - T = \frac{1}{(k + k_2 + \beta)^2}\left[(k + k_2 + \beta)^2 - (k - k_2 - \beta)^2 - 4kk_2\right]$

$= \frac{4k\beta}{(k + k_2 + \beta)^2}$

(1) $E<0$，束缚态，无节点 $\rightarrow$ 只能偶宇称 $\left.\frac{\partial\psi}{\partial x}\right|_{x=0}=0$

$x\neq\pm a$，$V(x)=0$ ：$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(x)=E\psi(x)$

解得 $\psi(x)=\begin{cases}Ae^{kx}+Be^{-kx} & x\in(0,a)\\ Ce^{-kx} & x\in(a,\infty)\end{cases}$　$k=\sqrt{-\frac{2mE}{\hbar^2}}$

偶宇称：$A=B$

$$\begin{cases}\psi(a)=\psi(a)\\ \frac{d}{dx}\psi|_{x=a^+}-\frac{d}{dx}\psi|_{x=a^-}=\frac{2m\gamma}{\hbar^2}V_0\psi(a)\end{cases}$$

$$\begin{cases}2A\cosh(ka)=Ce^{-ka}\\ -kCe^{-ka}-2kA\sinh(ka)=\frac{2m\gamma}{\hbar^2}Ce^{-ka}\end{cases}$$

$\Rightarrow k\tanh(ka)=k+\frac{2m\gamma}{\hbar^2}$

$\sinh x=\frac{1}{2}(e^x-e^{-x})$　$\cosh x=\frac{1}{2}(e^x+e^{-x})$

$k\cdot\frac{\sinh(ka)-\cosh(ka)}{\cosh(ka)}=\frac{ke^{-ka}}{e^{ka}+e^{-ka}}=\frac{2m\gamma}{\hbar^2}$

最小的为 $k_0$，基态能量 $E_0=-\frac{\hbar^2k_0^2}{2m}$

(2) $E>0$

$$\psi(x)=\begin{cases}Ae^{ikx}+Be^{-ikx}\\ Ce^{ikx}+De^{-ikx}\\ Ee^{ikx}\end{cases}$$

$$\begin{cases}Ae^{-ika}+Be^{ika}=Ce^{-ika}+De^{ika}\\ Ce^{ika}+De^{-ika}=Ee^{-ika}\\ ik(Ce^{-ika}-De^{ika}-Ae^{-ika}+Be^{ika})=\frac{2mV_0}{\hbar^2}(Ae^{-ika}+Be^{ika})\\ ik(Ee^{ika}-Ce^{ika}+De^{-ika})=\frac{2mV_0}{\hbar^2}Ee^{ika}\end{cases}$$